

How a Router Handles Responses for Multiple Devices in a Private Network

In a private network (like your home network), you can have multiple devices with **private IPs** such as `192.168.1.2`, `192.168.1.3`, etc. The router has **one public IP** for the internet. NAT (Network Address Translation), specifically **PAT (Port Address Translation)**, helps the router know which device a response belongs to.

Step 1: Device Sends a Request

Example: Laptop wants to access a website. - **Private IP/Port:** `192.168.1.2:54321` - **Destination IP/Port:** Website IP, port 80

The router replaces the **source IP** with its **public IP** and assigns a **unique public port** (e.g., `40000`).

NAT Table Entry: | Public Port | Private IP | Private Port | |-----|-----|-----| | 40000 |
192.168.1.2 | 54321 |

Step 2: Internet Server Responds

- The website sends a response to **public IP** and **port 40000**.
 - Router looks up **port 40000** in the NAT table.
 - Router finds it corresponds to `192.168.1.2:54321`.
 - Router rewrites destination IP/port to the private IP and forwards it to the correct device.
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Step 3: Another Device Sends a Request

Example: Second device (`192.168.1.3:5173`) sends a request. - Router assigns **public port 40001**

Updated NAT Table: | Public Port | Private IP | Private Port | |-----|-----|-----| | 40000 |
192.168.1.2 | 54321 | | 40001 | 192.168.1.3 | 5173 |

- Internet server responds to **public IP:40001**
 - Router looks up **port 40001** and forwards the response to `192.168.1.3:5173`
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Key Points

1. **Unique public port per connection** ensures multiple devices can access the internet simultaneously.
2. Router uses **public IP + port mapping** to know which private device should receive a response.
3. Private IPs are not visible to the internet; the NAT table handles all translations.

This is how a single public IP can serve multiple devices in a private network efficiently.